Akshay Patil

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EDUCATION

Georgia Institute of Technology – Atlanta, GA **August 2026 – May 2029**
Bachelor of Science in Aerospace Engineering
Current coursework: CS 1371 Computing, MATH 2551 Multivariable Calculus, COE 2001 Statics, ME 1670 Graphics, AE 1601 Aerospace Engineering, MSE 2001 Materials

School of Science and Engineering at Townview – Dallas, TX **August 2022 – May 2026**
3.97 Unweighted GPA | Rank 2 | 1600 SAT | National Merit Scholar | NASA High School Aerospace Scholar

RESEARCH

IVALab, Georgia Tech | Undergraduate Researcher **September 2026 – Present**

- The lab works on navigation for robots that have to move through crowds of people
- Developing navigation algorithms that mix learned policies with classical planners
- Benchmarking them against each other in the Arena simulator

EXPERIENCE

Hardpoint Robotics | Co-Founder **June 2026 – Present**

- Hardpoint builds NDAA-compliant development UAVs, aircraft for customers who are not allowed to fly foreign-sourced hardware
- I own the airframe: CAD, 3D printed CF-Nylon parts, and carbon fiber structures made with outside vendors
- Designing around sourcing as a hard constraint, since every part has to be compliant and purchasable, so vendor lead times shape the design as much as the loads do

Salvo Dynamics | Design Engineer **August 2026 – Present**

- Salvo builds small interceptor aircraft meant to fly out and bring down hostile drones, and is separately building an RC jet to break the speed of sound
- Turning that jet's aero body into a load-bearing structure in CAD and laying out the avionics that go inside it
- Designing a 3D printed airframe for testing, plus the assembly jigs the production aircraft are built on

SWYFT Robotics | Engineering Design Intern **May 2025 – May 2026**

- SWYFT sells competition drivetrain and structural parts to FIRST Tech Challenge and FIRST Robotics Competition teams
- Designed sprockets, gears, and pulleys that hold up under competition loads at a price teams can afford

PROJECTS

Autonomous House-Mapping Robot | Personal Project **2025 – Present**

- Jetson Orin Nano robot that explores a house with no operator and counts windows, using an MPPI controller, RTAB-Map visual SLAM, and frontier exploration I wrote
- No lidar: obstacles come from monocular depth classified against a fitted ground plane, windows from three open-vocabulary detectors filtered by a vision-language model

Pickleball Ball Boy UAV | Personal Project **May 2025 – Present**

- Benchmarked five ball detection and four localization methods; AprilTag localization was the most accurate by a wide margin
- Split the drag model into horizontal and vertical coefficients, bringing landing prediction inside a 20 cm zone

ENGINEERING TEAMS

FIRST Tech Challenge Team 7172 | Captain and Lead Designer **September 2021 – May 2026**

- Lead designer on four robots; 2026 World Championship Finalist Alliance, 19 event wins and 63 awards over five seasons
- Designed a shooter that sorts and fires three balls in 0.2 to 0.5 s using a wedge spindexer and a flywheel on a 200 Hz servo turret

Yellow Jacket Space Program | Turbomachinery, Engine Development **August 2026 – Present**

- The program is going for the collegiate liquid propellant altitude record and the Karman line at 100 km
- Designing electric pumps and turbopumps so the vehicle can drop the large, heavy pressurant tanks a pressure-fed engine needs

SKILLS

Robotics: ROS 2 (Nav2), RTAB-Map visual SLAM, MPPI control, AprilTag localization
Programming: Python (NumPy, Pandas, TensorFlow), MATLAB, JavaScript, SQL, Git
Hardware: NVIDIA Jetson, Raspberry Pi, Arduino, REV Control Hub
CAD: SolidWorks, Fusion 360, Onshape, Autodesk Inventor, 3D printing
Certifications: FAA Part 107 Remote Pilot, Autodesk Certified User (Inventor)